

Question:

Putting into consideration a market vendor, with examples, discuss the concept of functions.

In programming, a function is just like a job or task a market vendor repeats over and over. You give it inputs, it does work and gives you an output.

1. What is a function?

function = a reusable block of code that does one specific task, for example;

Think of a market vendor who sells matooke:

Programming term	Market vendor example
Function name	Weigh_matooke()
Input/ Parameter	The bunch of matooke the customer brings
Work inside function	Put it on the scale, calculate price
Output/ Return	That's 3kg. Pay 6,000 UGX

Instead, of doing the weighing math every time, the vendor has a set “procedure” they run for every customer.

2. Why functions matter

2.1. Reusability: Write once, use it many times. The vendor doesn't invent a new way to weigh for each customer.

2.2. Less repetition: there's no copying the same code everywhere.

2.3. Easier to fix: if the scale is wrong, you fix the “weigh_matooke ()” function once.

3. Python example: Market vendor's procedure;

Define a function - the vendor's procedure

```
def sell_matoke(kg, price_per_kg): total = kg * price_per_kg # do the  
calculation return total # give back the result
```

Use the function - different customers

```
customer1 = sell_matoke(3, 2000) # 3kg at 2000 UGX/kg
```

```
customer2 = sell_matoke(5, 2000) # 5kg at 2000 UGX/kg
```

```
customer3 = sell_matoke(1.5, 2000) # 1.5kg at 2000 UGX/kg
```

```
print(f"Customer 1 pays: {customer1} UGX") # 6000 UGX  
print(f"Customer 2 pays: {customer2} UGX") # 10000 UGX  
print(f"Customer 3 pays: {customer3} UGX") # 3000 UGX
```

Breaking it down:

1. `def sell_matooke (kg, price_per_kg):` this is defining the vendor's procedure. Kg and price_per_kg are inputs.
2. `total = kg * price_per_kg :` this is work done inside.
3. `Return total:` the vendor hands you the total cost. That's the output.

4. Another example: Giving change function;

```
def give_change(money_given, total_cost):
```

```
    change = money_given - total_cost
```

```
    return change
```

```
print(give_change(10000, 6000)) # Output: 4000
```

Note:

bottom line; functions = named tasks you can call anytime. Just like a vendor has standard tasks: weigh, calculate price, give change.